MODE OF RESPIRATION IN ANIMALS AND PLANTS WITH EXAMPLES

Learning objectives

By the end of this unit, each student will be able to:

1. define respiration and the various types of respiration
2. state the various modes of respiration seen in animals and plants
3. list organisms that respire through the various modes

General Overview

Organisms carry out important life processes like synthesis of proteins, fats, carbohydrates and other metabolic reactions. Energy is required for these activities. One might therefore aptly ask: where does the energy come from, how is it stored and how is it made available to the cells of these organisms?

During the process of photosynthesis, light energy is converted to chemical energy and stored in the bonds of complex organic molecules - in the form of carbohydrates, such as starch and glucose. The weakening or breaking of the carbon to carbon bonds of such compounds releases a considerable amount of energy for utilization by the plants. However, the total amount of energy contained within a compound for example, glucose, is not released all at once, but slowly, in a stepwise series of reactions controlled by enzymes.

Generally series of reactions in the cell leading to the synthesis or breakdown of organic compound is referred to as a metabolic pathway. In plants, there are two chemical processes occurring at every particular point in time. These are the building up (synthesis process) called anabolism e.g. photosynthesis in plants and a breakdown process called catabolism e.g. respiration. In animals, respiration is catabolic.

Respiration

Carbohydrates produced in photosynthesis serve as ready source of stored energy and are in most cases preferentially utilized as respiratory substrates. The energy is trapped in high energy compounds; the most important among them is the ATP; NADH and NADPH are equally important because of their ability to transfer electrons. The chains of oxidation-reductions reactions involved are collectively referred to as respiration**.**

Respiration as an oxidation process is a chain of chemical reactions involving the degradation of complex organic compound, from which carbon dioxide, water and energy are produced.

Types of Respiration

Respiration is of two types:

1. Aerobic Respiration

Aerobic respiration is a series of enzyme-controlled reactions that release the energy stored up in carbohydrates and lipids during photosynthesis and make it available to living organisms.

In most animals, the tissue oxidation occurs in presence of O_2 with end products CO_2 and H_2O. It requires exchange of O_2 and CO_2 between the organisms and the surroundings. Aerobic respiration takes place inside mitochondria. The pathways involved include Glycolysis, Krebs cycle and Electron transport system as sketchily represented in Fig. 1 below

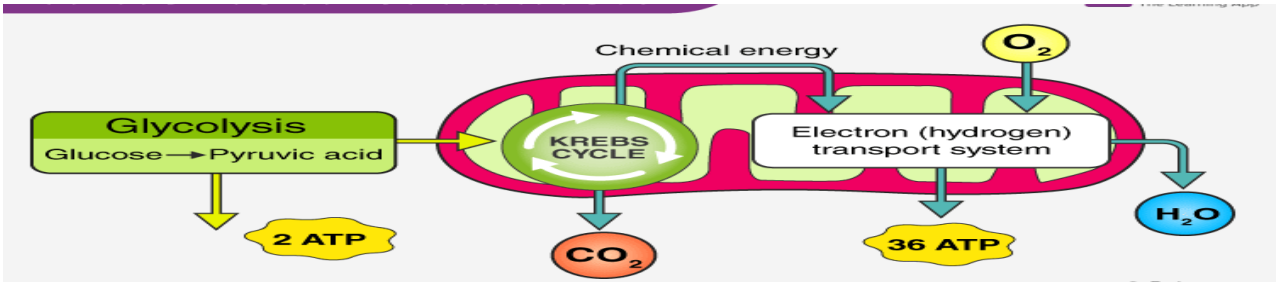


Fig.1. Pathways of Aerobic Respiration in Organisms

Aerobic respiration may be represented by the general equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

Anaerobic Respiration

In lower organisms like bacteria, yeast, and so on, the nutrients are oxidized without using O_2 and lactic acid or ethyl alcohol is the end product. It also occurs in some multicellular animals and some tissues (skeletal muscle and RBC) of higher animals. Anaerobic respiration occurs outside mitochondria of cells.

During glycolysis, glucose molecules (six-carbon molecules) are split into two pyruvates (three-carbon molecules) during a sequence of enzyme-controlled reactions. This is the same reaction as occurs in aerobic respiration. Without oxygen, pyruvate is converted to lactic acid in animals or ethanol in plants and yeast. It produces only about 10% of the energy released in the complete oxidation of glucose. Anaerobic respiration in humans takes place when muscle undergoes extreme contraction as in vigorous exercise. When oxygen is limited the oxidation of NADH to NAD^+ by the electron transport chain is insufficient to maintain glycolysis. Under these conditions NAD^+ is regenerated by the reduction of pyruvate to lactate.

In yeast, pyruvate is converted to ethanal and then to ethanol. The latter stage oxidizes NADH to NAD^+, allowing glycolysis to continue.

Differences between Aerobic and Anaerobic Respiration

	Aerobic respiration	**Anaerobic respiration**
Usage of Oxygen	It uses molecular oxygen	It does not use molecular oxygen.
Release of CO_2	Always release CO_2.	May or may not release CO_2.
Production of water	It produces water.	It does not produce water
Quantity of energy produced	It produce much more energy (whole energy present in glucose)	It produces less energy (only 5% of that available in glucose).
End product	It yields inorganic end products only.	It yields organic end products with or without inorganic product.

Respiration process can be divided into two phases:

1. **External Respiration:** In this process, animals take in oxygen and release carbon dioxide. It is also known as breathing. This is mainly two phases.

a) Inspiration: It is the process by means of which oxygen is taken to the lungs.

b) Expiration: During expiration, carbon dioxide is expelled out from the respiratory organs into the environment.

2. **Internal Respiration or Cellular Respiration:** In this process, oxygen is used in chemical reactions within the cells. These reactions release energy from food substances and produce carbon dioxide and water as waste products. Sugar + $O_2 \longrightarrow CO_2 + H_2O$ + Energy.

Mode of Respiration in Animals

The mode of external respiration varies greatly from organism to organism. Organisms adopt different modes of respiration depending on their types, complexity, size and the habitat in which they are found. Respiratory structures are also tailored to the organism's need for oxygen. Thus, below are respiratory structures and mechanisms used by various organisms:

1. Diffusion through Cell surface/Plasma Membrane/Body Surface

In unicellular animals, such as amoeba, exchange of gases takes place through the cell surface. They absorb oxygen from the surrounding air or water and give out carbon dioxide by diffusion.

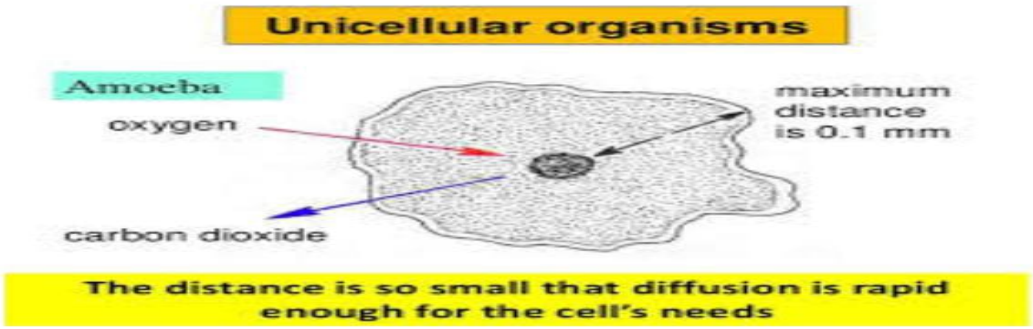


Fig.2: Diffusion in Amoeba

Organism like tapeworms, earthworms, and leeches also use their skin for the exchange of gases.



Fig. 3: Respiration in Earthworm

The skin of earthworms is very thin and moistened. Many blood cells are spread on this skin. These blood cells are known as capillaries. The exchange of gases occurs at capillaries. Oxygen needed for respiration is dissolved in water. On the principle that molecular gases diffuse from a region of higher concentration to region of lower concentration, dissolved oxygen which is of higher concentration in water than in the cell will diffuse into the cell. After gaseous exchanges, carbon(iv)oxide which is now higher in concentration in the cell than in water will diffuse out of the cell into the water. They die of suffocation if their skin is dried up.

2. Through the Trachea

Trachea is a common respiratory organ of invertebrates. This respiratory organ is a hallmark of insects like cockroaches, grasshopper e.t.c. It is made up of a system of branching tubes that deliver

oxygen to, and remove carbon dioxide from the tissues. The pores to the outside called spiracles are typically paired structures, two in the thorax and eight in the abdomen. Periodic opening and closing of the spiracles prevents water loss by evaporation, a serious threat to insects that live in dry environments.

Oxygen for respiration enters the abdomen and thoracic segment from surrounding atmosphere. The expansion of the abdomen causes the volume of the abdomen to increase thereby drawing air by diffusion into the body through the spiracles, tracheae, air sacs and trachioles. When the atmospheric oxygen reaches the fluid of the trachioles, it dissolves. The dissolved oxygen then diffuses into the tissues where cellular or tissue respiration occurs. When the thoracic segments and abdomen contract, carbon (iv)oxide and water vapour diffuse from spiracles into the atmosphere.

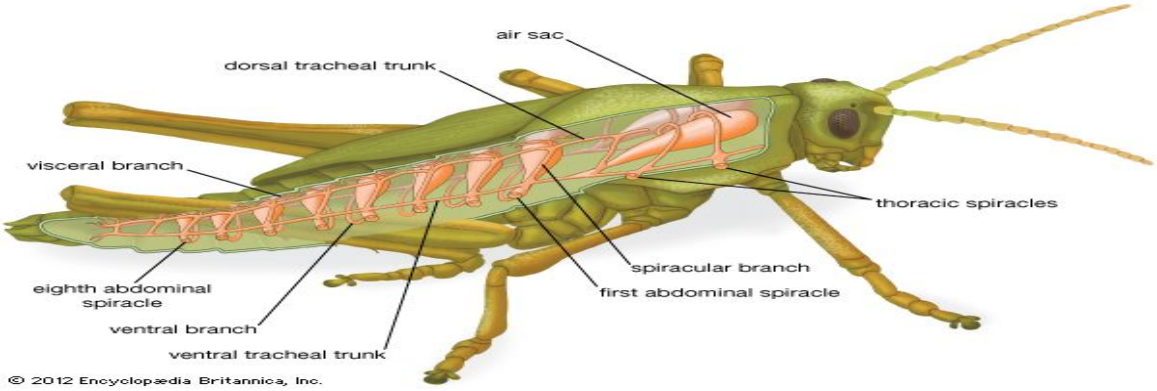


Fig 4: Tracheal System in Grasshopper

Although tracheal systems are primarily designed for life in air, in some insects, **modifications enable the tracheae to serve for gas exchange under water**. Of special interest are the insects that might be termed **bubble breathers,** which, as in the case of the water beetle ***Dytiscus***, take on a gas supply in the form of an air bubble under their wing surfaces next to the spiracles before they submerge. Tracheal gas exchange continues after the beetle submerges and anchors beneath the surface. As oxygen is consumed from the bubble, the partial pressure of oxygen within the bubble falls below that in the water; consequently oxygen diffuses from the water into the bubble to replace that consumed. The carbon dioxide produced by the insect diffuses through the tracheal system into the bubble and thence into the water. The bubble thus behaves like a **gill**. There is one major limitation to this adaptation: As oxygen is removed from the bubble, the partial pressure of the nitrogen rises, and this gas then diffuses outward into the water. The consequence of outward nitrogen diffusion is that the bubble shrinks and its oxygen content must be replenished by another trip to the surface.

A partial solution to the problem of bubble renewal has been found by small aquatic beetles of the family Elmidae (*e.g., **Elmis***, *Riolus*), which capture **bubbles containing oxygen produced by algae and incorporate this gas into the bubble gill**. Several species of aquatic beetles also augment gas exchange by stirring the surrounding water with their posterior legs.

One extraordinary strategy used by the hemipteran insects *Buenoa* and *Anisops* is an **internal oxygen store that enables them to lurk for minutes without resurfacing** while awaiting food in relatively predator-free but oxygen-poor mid-water zones. The internal oxygen store is in the form of hemoglobin-filled cells that constitute the first line of oxygen delivery to actively metabolizing cells, sparing the small air mass in the tracheal system while the hemoglobin store is being depleted.

Immature insects - nymphs

Many immature insects have special adaptations for an aquatic existence. Thin-walled protrusions of the integument, containing tracheal networks, form a series of gills (tracheal gills) that bring water

into close contact with the closed tracheal tubes. The nymphs of mayflies and dragonflies have external tracheal gills attached to their abdominal segments, and certain of the gill plates may move in a way that sets up water currents over the exchange surfaces. Dragonfly nymphs possess a series of tracheal gills enclosed within the rectum. Periodic pumping of the rectal chamber serves to renew water flow over the gills. Removing the gills or plugging the rectum results in lower oxygen consumption. Considerable gas exchange also occurs across the general body surface in immature aquatic insects.

NB: The insect tracheal system has inherent limitations. Gases diffuse slowly in long narrow tubes, and effective gas transport can occur only if the tubes do not exceed a certain length. It is generally thought that this has imposed a size limit upon insects.

3. Through Book lung

The respiratory structures of spiders consist of peculiar "book lungs," leaflike plates over which air circulates through slits on the abdomen. The book lungs contain blood vessels that bring the blood into close contact with the surface exposed to the air and where gas exchange between blood and air occurs. In addition to these structures, there may also be abdominal spiracles and a tracheal system like that of insects.

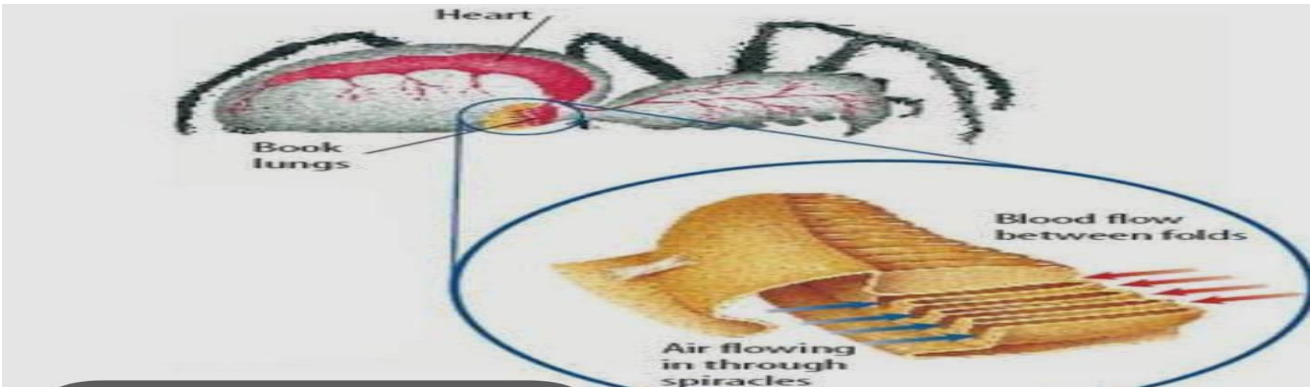


Fig. 5: Book lung of a Spider

Since spiders are air breathers, they are mostly restricted to terrestrial situations, although some of them regularly hunt aquatic creatures at stream or pond edges and may actually travel about on the surface film as easily as on land. The water spider (or diving bell spider), *Argyroneta aquatic* - known for **its underwater silk web**, which resembles a kind of diving bell - is the only species of spider that spends its entire life underwater. Using **fine hairs on its abdomen,** where its respiratory openings lie, the water spider captures tiny bubbles of air at the water's surface, transports them to its silk web, which is anchored to underwater plants or other objects, and ejects them into the interior, thereby inflating the underwater house with air. Research has shown that the inflated web serves as a sort of gill, extracting dissolved oxygen from the water when oxygen concentrations inside the web become sufficiently low to draw oxygen in from the water. As the spider consumes oxygen, nitrogen concentrations in the inflated web rises, causing it to slowly collapse. Hence, the spider must travel to the water's surface for bubble renewal, which it does about once each day. Most of the life cycle of the water spider, including courtship and breeding, prey capture and feeding, and the development of eggs and embryos, occurs below the water surface. Many of these activities take place within the spider's diving bell.

4. Gills of Invertebrates

Gills are evaginations of body surface. Some open directly to the environment; others, as in fishes, are enclosed in a cavity just as lungs represent invaginations of the body surface. Many invertebrates use gills as a major means of gas exchange; a few, such as the pulmonate land snail, use lungs. Almost any thin-walled extension of the body surface that comes in contact with the environmental medium and across which gas exchange occurs can be viewed as a gill. Gills usually have a large

surface area in relation to their mass; pumping devices are often employed to renew the external medium. Although gills are generally used for water breathing and lungs for air breathing, this association is not invariable, as exemplified by the water lungs of sea cucumbers.
The marine polychaete worms use not only the general body surface for gas exchange but also a variety of gill-like structures: segmental flaplike parapodia (in *Nereis*) or elaborate branchial tufts (among the families Terebellidae and Sabellidae). The tufts, used to create both feeding and respiratory currents, offer a large surface area for gas exchange.

In echinoderms (starfish, sea urchins, brittle stars), most of the respiratory exchange occurs across tube feet (a series of suction-cup extensions used for locomotion). However, this exchange is supplemented by extensions of the coelomic, or body-fluid, cavity into thin-walled "gills" or dermal branchiae that bring the coelomic fluid into close contact with seawater. Sea cucumbers (Holothuroidea), soft-bodied, sausage-shaped echinoderms that carry on some respiration through their oral tentacles, which correspond to tube feet, also have an elaborate "respiratory tree" consisting of branched hollow outpouchings off the cloaca (hindgut). Water is pumped in and out of this system by the action of the muscular cloaca, and it is probable that a large fraction of the animals' respiratory gas is exchanged across this system.

The gills of mollusks have a relatively elaborate blood supply, although respiration also occurs across the mantle, or general epidermis. Clams possess gills across which water circulates, impelled by the movements of millions of microscopic whips called cilia. In the few forms studied, the extraction of oxygen from the water has been found to be low, on the order of 2 to 10 percent. The currents produced by cilial movement, which constitute ventilation, are also utilized for bringing in and extracting food. At low tide or during a dry period, clams and mussels close their shells and thus prevent dehydration. Metabolism then shifts from oxygen-consuming (aerobic) pathways to oxygen-free (anaerobic) pathways, which causes acid products to accumulate; when normal conditions are restored, the animals increase their ventilation and oxygen extraction in order to rid themselves of the acid products. In snails, the feeding mechanism is independent of the respiratory surface. A portion of the mantle cavity in the form of a gill or "lung" serves as a gas-exchange site. In air-breathing snails, the "lung" may be protected from drying out through contact with the air by having only a pore in the mantle as an opening to the outside. Cephalopod mollusks, such as squid and octopus, actively ventilate a protected chamber lined with feathery gills that contain small blood vessels (capillaries); their gills are quite effective, extracting 60 to 80 percent of the oxygen passing through the chamber. In oxygen-poor water, the octopus may increase its ventilation 10-fold, indicating a more active control of respiration than appears to be present in other classes of mollusks. Many crustaceans (crabs, shrimps, crayfish) are very dependent on their gills.

Respiratory Organs of Vertebrates

In most vertebrates, the organs of external respiration are thin-walled structures well supplied with blood vessels. Such structures bring blood into close association with the external medium so that the exchange of gases takes place across relatively small distances. There are three major types of respiratory structures in the vertebrates: a) **gills**, b) **integumentary exchange areas**, and c) **lungs**. The gills are totally external in a few forms (as in *Necturus*, a neotenic salamander), but in most they are composed of filamentous leaflets protected by bony plates (as in fish). Some fishes and numerous amphibians also use the body integument, or skin, as a gas-exchange structure. Both gills and lungs are formed from outpouchings of the gut wall during embryogenesis. Such structures have the advantage of a protected internal location, but this requires some sort of pumping mechanism to move the external gas-containing medium in and out.
The quantity of air or water passing through the lungs or gills each minute is known as the **ventilation volume**. The maximum capacity of human lungs is about six litres. During normal

quiet respiration, a tidal volume of about 500 millilitres is inspired and expired during every respiratory cycle. The lungs are not collapsed at the close of expiration; a certain volume of gas remains within them. At the close of the expiratory act, a normal subject may, by additional effort, expel another 1,200 millilitres of gas. Even after the most forceful expiratory effort, however, there remains a residual volume of approximately 1,200 millilitres. By the same token, at the end of a normal inspiration, further effort may succeed in drawing into the lungs an additional 3,000 millilitres.

5. The Gills of Vertebrates

A majority of aquatic animals like fish and prawns breathe through special organs called gills. Gilla are projections of the skin that help in using oxygen dissolved in water. Gills contain blood vessels which help in exchange of gases.

Gill in Amphibians

Tadpole: This is a larval toad or frog that hatch from eggs and can only survive in water. Tadpoles have tiny external gill flaps that extract oxygen from water as it passes over them. Tadpoles open their mouths as they swim and take in water. As the mouth closes, muscles transfer the water to the gills. These consist of thin membranes called lamellae, which take oxygen from the water where it enters the blood stream through the process of diffusion. Tadpoles can also rise to the surface and gulp oxygen from the air. As tadpoles mature the gills are absorbed by the body as other respiratory systems develop. In addition to these structures, frog tadpoles use their large tail fins for respiration; the tail fins contain blood vessels and are important respiratory structures because of their large surface area. As amphibian larvae develop, the gills (and in frogs, the tail fin) degenerate, paired lungs develop, and the metamorphosing larvae begin making excursions to the water surface to take air breaths.

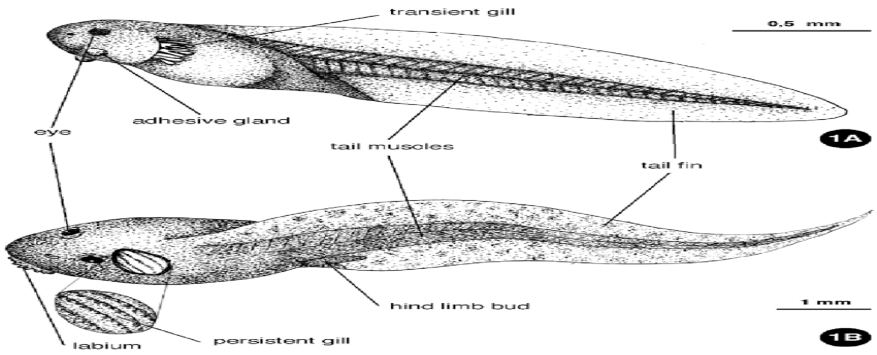


Fig.7 Gill respiration in tadpole

Adult Toad: The adult toad respire in three different ways namely: the mouth, skin and lungs

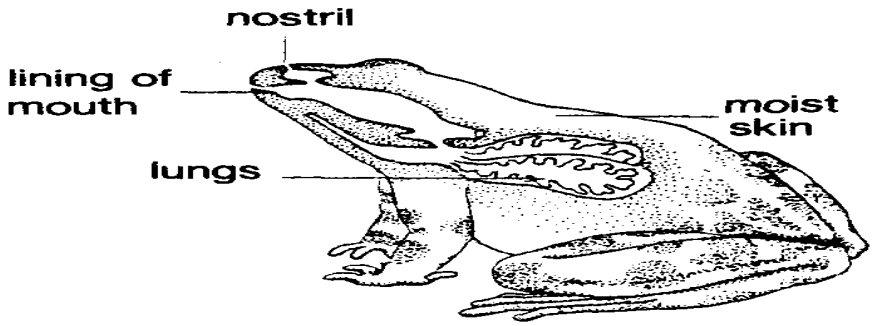
1. Buccal (Mouth) Gaseous Exchange

The toad is able to utilize its mouth as a respiratory organ because of the following:

- It is very large i.e. has a large surface area.
- It has a thin mucus membrane for easy diffusion
- It is well supplied with blood capillaries

To initiate breathing, the toad closes its mouth, the nostrils are opened and the floor of the buccal cavity is lowered, air is drawn through the nostrils into the buccal cavity. After this, the capillaries and the glottis close and gaseous exchange takes place between the blood and the inhaled air. To get rid

of air, the floor of the buccal cavity is raised increasing the air pressure, and hereby forcing the nostrils to open and air in the buccal cavity containing carbon dioxide flows out.



Breathing organs of the frog

2. Skin/Cutaneous Respiration: This is possible because of the large surface area of the skin. The skin is moist because of continuous secretion from the mucus gland, there is adequate supply of blood capillaries and finally, the membrane serving as the skin is thin. Simple diffusion of gases takes place through it both on land and water.

3. Lungs/Pulmonary Respiration in Toad

The trachea of amphibians is not divided into secondary tubes but ends abruptly at the lungs. The relatively simple lungs of frogs are subdivided by incomplete walls (septa), and between the larger septa are secondary septa that surround the air spaces where gas exchange occurs. The diameter of these air spaces (alveoli) in lower vertebrates is larger than in mammals: The alveolus in the frog is about 10 times the diameter of the human alveolus. In order to draw air into its mouth the toad lowers the floor of its mouth, which causes the throat to expand. Then the nostrils open allowing air to enter the enlarged mouth. The nostrils then close and the air in the mouth is forced into the lungs by contraction of the floor of the mouth. To eliminate the carbon dioxide in the lungs the floor of the mouth moves down, drawing the air out of the lungs and into the mouth. Finally the nostrils are opened and the floor of the mouth moved up pushing the air out of the nostrils.

The mechanism of lung inflation in amphibians is the buccal cavity (mouth-throat) pumping mechanism that also functions in air-breathing fishes. To produce inspiration, the floor of the mouth is depressed, causing air to be drawn into the buccal cavity through the nostrils. The nostrils are then closed, and the floor of the mouth is elevated. This creates a positive pressure in the mouth cavity and drives air into the lungs through the open glottis. Expiration is produced by contraction of the muscles of the body wall and the elastic recoil of the lungs, both acting to drive gas out of the lungs through the open glottis. In aquatic amphibians the pressure of water on the body wall can also assist expiration. Many amphibians show rhythmic oscillations of the floor of the mouth between periods of lung inflation; these oscillations are thought to be involved in olfaction by producing a flow of gas over the olfactory epithelial surfaces.

Gills in Fish living in Water

The gills of fishes are supported by a series of gill arches encased within a chamber formed by bony plates (the operculum). A pair of gill filaments projects from each arch; between the dorsal (upper) and ventral (lower) surfaces of the filaments, there is a series of secondary folds, the lamellae, where the gas exchange takes place. Deoxygenated blood from the heart flows in the lamellae in a direction counter to that of the water flow across the exchange surfaces. In a number of fishes the water-to-blood distance across which gases must diffuse is 0.0003 to 0.003 millimeter, or about the same distance as the air-to-blood pathway in the mammalian lung.

The countercurrent flow of blood through the lamellae in relation to external water flow has much to do with the efficiency of gas exchange. Laboratory experiments in which the direction of water flow across fish gills was reversed showed that about 80 percent of the oxygen was extracted in the normal situation, while only 10 percent was extracted when water flow was reversed. Countercurrent flow is a feature of elasmobranchs (sharks, skates) and cyclostomes (hagfishes, lampreys) as well as bony fishes.

To get oxygen dissolved in water, fish gulp water through the mouth and pump it over the gills. Water passes into the gill chamber through gill slits. In each chamber, the water passes over the filaments. They absorb oxygen from the water and replace it with carbon dioxide formed. The water then passes out through the gill opening and this process is repeated again and again. Among the most primitive of present-day vertebrates are the cyclostomes (lampreys and hagfishes), the gill structures of which are in the form of pouches that connect internally with the pharynx (throat) and open outward through slits, either by a fusion of the excurrent gill ducts into a single tube (in *Myxine*) or individually by separate gill slits (in *Petromyzon*). The gill lamellae of cyclostomes form a ring around the margins of the gill sac, and the series of sacs is supported in a flexible branchial skeleton. The number of paired pouches varies in different forms from six to 14. The pharynx of lampreys divides into an esophagus above and a blind tube below, from which the gill pouches arise. The upper pharynx of hagfishes communicates to the exterior through a nostril, a structure absent in lampreys. When the parasitic lampreys are embedded in the flesh of fish, upon which they live, they maintain a flow of water through the gills by alternate contractions of the gill pouches. When the gill-pouch muscles relax, the pouches expand, and water is sucked in. The water is forced out through the gills by muscular contraction; the branchial musculature apparently prevents reflux of the water into the pharynx while the head of the lamprey is embedded in the flesh of its prey.

A number of vertebrates use externalized gill structures. Some larval fishes have external gills that are lost with the appearance of the adult structures. A curious example of external gills is found in the male lungfish (*Lepidosiren*). At the time the male begins to care for the nest, a mass of vascular filaments (a system of blood vessels) develops as an outgrowth of the pelvic fins. The fish meets its own needs by refilling its lungs with air during periodic excursions to the water surface. When it returns to the nest, its pelvic-gill filaments are perfused with well-oxygenated blood, providing an oxygen supply for the eggs, which are more or less enveloped by the gill filaments.

In the hagfish *Myxine glutinosa*, the major oxygen supply is derived from water drawn in through the nostril that opens into the pharynx. A peculiar respiratory structure, the velum, just behind the nostril opening, dangles from the upper midline of the pharynx, resembling an inverted T. Membranous scrolls attached to this horizontal bar can extend downward and then roll upward like window shades. A combination of velar and gill-pouch contractions directs the flow of water through the gill pouches. Foreign material entering the nostril is expelled from both the mouth and nostril by a violent “sneeze.” This reaction probably protects the respiratory surfaces, since the animals have common respiratory and alimentary ducts. Blood flow in the gills of cyclostomes, as in those of bony fishes, is in a direction counter to that of water flow—an arrangement that increases the efficiency of gas exchange across the respiratory surface.

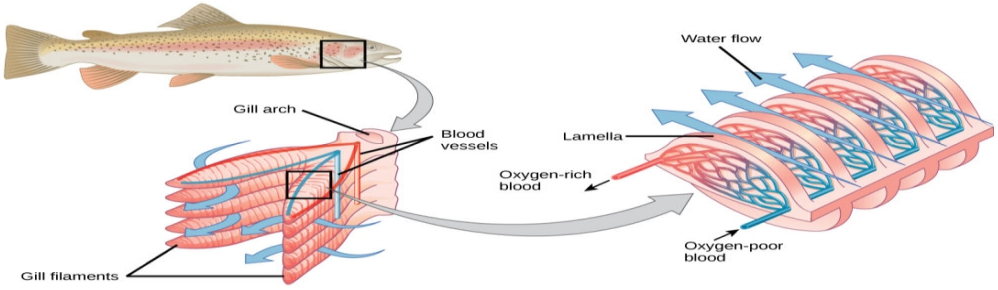


Fig. 8: Gills respiration in fish

Cartilaginous fishes (sharks and rays) and bony fishes employ a double-pumping mechanism to maintain a relatively constant flow of water over the gill exchange surfaces. In sharks and rays a small forward gill slit, the spiracle, also provides a channel for water flow into the gill chamber. Bottom-dwelling forms (*e.g.,* skates) have relatively larger spiracles, and the major portion of the water flow passes through them rather than through the downward-oriented mouth.

The pumping mechanism is not the only method of ventilation; sharks have been observed to keep both mouth and gill flaps open while swimming, ensuring a constant water flow across the gill surfaces. When they slow down or settle to the bottom, the pumping activity is resumed. Tunas and mackerel cannot stop swimming: They have no active respiratory mechanism and are dependent for their gill ventilation on the current that results from their forward motion through the water.

A number of fishes depend in varying degree on aerial respiration. The ability to breathe air enables them to live in places where the oxygen content of water may be low or nil. Two general means of acquiring oxygen are employed. Some fishes stay near the surface of the water, where the oxygen pressure resulting from surface diffusion is highest. Others have developed ancillary respiratory structures in the pharynx or the stomach; the gulping of air at the surface is a means of charging these respiratory surfaces (such as the pharyngeal epithelium in *Electrophorus* or the stomach in *Plecostomus*). The frequency with which these fishes rise to the surface to gulp air corresponds to their current need for oxygen.

The swamp-dwelling *Erythrinus* of Guyana uses both aquatic and aerial respiration, varying them according to the gaseous composition of the water. When the oxygen content is low, respiration through the gills ceases; when the oxygen content of the water is high, the fish relies primarily upon its gills except when the carbon dioxide content is also high—when, again, aerial respiration predominates. In other conditions it uses both modes of respiration. This apparently extends the range of conditions in which *Erythrinus* can survive.

Eels (*Anguilla*) use their skin as a major respiratory surface in addition to their gills. In water, about 15 percent of their oxygen uptake is across the skin, and this rises to around 50 percent when in air. They are capable of making extensive overland migrations during which, in the first few hours, they draw upon oxygen in the swim bladder. Like most fishes, eels when out of water exhibit a reduced heart rate and less oxygen consumption. When they return to water, their heart rate rises, and both oxygen consumption and blood lactic-acid levels rise. Lactic-acid production results from metabolism without oxygen, and such acid products must themselves be metabolized through higher oxygen consumption. Such patterns have been observed in grunions of the California coast that come ashore to breed, and even in flying fishes during their brief aerial excursions.

The lungfishes (Dipnoi) are remnants of the Devonian period and a transitional form between water and air breathers. Like amphibians, they rely on the buccal force pump mechanism to inflate the lung. They are adapted for bimodal respiration so that oxygenated blood leaving the air-exchange organ, either gills or lung, can pass to the afferent branchial circulation and then to the body tissues or can be dispatched to the lungs. The three dipnoan genera differ with respect to their reliance on the gills or lungs. In the Australian lungfish (*Neoceratodus*), the bulk of oxygen uptake and carbon dioxide elimination is by way of well-developed gills; in the African lungfish (*Protopterus*) and the South American lungfish (*Lepidosiren*), the gills are reduced and ventilation depends heavily on the lungs. In the latter two, which rely primarily on lung ventilation, separation of oxygenated and deoxygenated blood is much more complete than in the Australian lungfish.

During long periods of drought, both *Protopterus* and *Lepidosiren* build a subterranean cocoon that opens to the surface via a thin tunnel. They then enter into a state of estivation in which metabolism, respiration, and heart rate fall to low levels. This state of diminished oxygen requirement enables the lungfish to remain viable without food or water for months or years, until the waters return.

The bowfin, *Amia calva*, has both gills and an air bladder that may be used for respiration. It is almost exclusively a water breather at 10° C (50° F), a temperature at which it shows low physical activity. Its air-breathing rate increases with temperature and activity, and, at around 30°C (86°F), it draws about three times as much oxygen from air as from water. As in lungfishes, carbon dioxide elimination is predominantly across the gills. The bowfin's air-breathing frequency varies inversely with the oxygen content of the water; when oxygen tensions in water decline below 40 or 50 millimetres of mercury at 20°C (68°F), air breathing largely replaces water breathing. When an exchange surface (gill or air bladder) is not being utilized as the primary oxygen-exchange site, there is a tendency for blood to bypass it.

The so-called electric eel of South America (*Electrophorus electricus*) inhabits muddy streams that may become severely oxygen deficient. It is an obligatory air breather that depends upon the exchange of oxygen across the membranes of its mouth, expelling the air through its gill slits. Its blood has a high percentage of red corpuscles, is high in hemoglobin, and has an oxygen-absorbing capacity similar to that of mammals. Carbon dioxide elimination is primarily across the skin and, to a lesser extent, through the vestigial gills.

6. Through the Lungs

Amphibians, reptiles, birds and mammals exchange gases through special respiratory organs called lungs. Lungs are air-filled sac-like structures in the chest cavity. They are connected to the outside by a series of tubes and small opening. An important characteristic of lungs is their elasticity. An elastic material is one that tends to return to its initial state after the removal of a deforming force. Elastic tissues behave like springs. As the lungs are inflated, there is an accompanying increase in the energy stored within the elastic tissues of the lungs, just as energy is stored in a stretched rubber band. The conversion of this stored, or potential, energy into kinetic, or active, energy during the deflation process supplies part of the force needed for the expulsion of gases. A portion of the energy put into expansion is thus recovered during deflation. The elastic properties of the lungs have been studied by inflating them with air or liquid and measuring the resulting pressures. Muscular effort supplies the motive power for expanding the lungs, and this is translated into the pressure required to produce lung inflation. It must be great enough to overcome (1) the elasticity of the lung and its surface lining; (2) the frictional resistance of the lungs; (3) the elasticity of the thorax or thoraco-abdominal cavity; (4) frictional resistance in the body-wall structures; (5) resistance inherent in the contracting muscles; and (6) the airway resistance. The laboured breathing of the asthmatic is an example of the

added muscular effort necessary to achieve adequate lung inflation when airway resistance is high, owing to narrowing of the tubes of the airways.

Lungs in Reptiles

To survive on land, the reptiles had to develop a skin relatively impermeable to water, so as to prevent desiccation, and hence not well suited for respiration. Thus, while a few specialized reptiles (for example, sea snakes) can acquire nearly half of their oxygen supply through their skin, most reptiles depend almost entirely on the lungs for gas exchange. Reptilian lungs are considerably more complex than those of amphibians, showing much more internal partitioning to provide additional surface area for gas exchange between lung gas and blood. The most complex reptilian lungs are found in sea turtles such as *Chelonia mydas*, the green turtle. This species can develop a high metabolic rate associated with its prolific swimming ability. Its lungs are suited to providing a high rate of gas exchange, with extensive branching of the airways leading to the numerous gas sacs of the lungs.

The mechanism for lung inflation in reptiles is an aspiration (suction) pump, which is the same in general principle as the lung inflation mechanism in birds and mammals. In most reptiles inspiration is produced by muscular expansion of the rib cage and body wall, creating a sub-atmospheric pressure within the lungs that causes air to flow in. Crocodiles and alligators have a specialized muscle attached to the posterior surface of the liver; the anterior surface of the liver in turn is attached to the posterior surface of the lungs. Contraction of this muscle pulls on the liver and results in expansion of the lungs.

The adoption of a rigid shell by turtles and tortoises necessitated the development of highly specialized skeletal muscles to inflate the lungs. In the tortoise *Testudo graeca*, lung ventilation is achieved by changing the volume of the body cavity. Expiration is brought about by the activity of muscles that draw the shoulder girdle back into the shell, compressing the abdominal viscera. The increased pressure in the body cavity is transmitted to the lungs. Inspiration involves opposite muscular actions that produce an increase in the volume of the body cavity and thus a subatmospheric lung pressure. Because of the rigidity of its shell, the tortoise, unlike other reptiles, cannot use the potential energy of abdominal wall structures to assist in respiration, and hence both expiration and inspiration are active energy-consuming events. In aquatic turtles, however, the pressure of water on the front and rear limbs assist in expiration.

The breathing patterns of most reptiles are not regular, usually consisting of a series of active inspirations and expirations followed by **relatively long pauses.** In aquatic reptiles diving occurs during these pauses, which may last an hour or more in some turtles and aquatic snakes. Even terrestrial reptiles show intermittent periods of breathing and breath-holding. The metabolic rate of most reptiles is one-fifth to one-tenth that of birds or mammals, and constant lung ventilation is unnecessary in most reptiles.

Lungs in Birds

Birds must be capable of high rates of gas exchange because their oxygen consumption at rest is higher than that of all other vertebrates, including mammals, and it increases many times during flight. The gas volume of the bird lung is small compared with that of mammals, but the lung is connected to voluminous air sacs by a series of tubes, making the total volume of the respiratory system about twice that of mammals of comparable size. The trachea divides into primary bronchi, each of which passes through a lung and onward to the paired abdominal air sacs; they also give rise to secondary bronchi supplying the other air sacs. Tertiary bronchi penetrate the lung mass and, from the walls of the tertiary bronchi, rather fine air capillaries arise. These air capillaries have a large surface area; their walls contain blood capillaries connected with the heart. Gas exchange takes place

between the air capillaries and blood capillaries, making this surface analogous to the alveolar surface in mammals.

There are several important differences in the mechanism and pattern of lung ventilation in birds compared with other vertebrates with lungs. The lungs of birds do not inflate and deflate but rather retain a constant volume. Also, the lungs are unidirectionally ventilated rather than having a tidal, bidirectional flow, as in other vertebrates with lungs. To achieve this unidirectional flow, the various air sacs are inflated and deflated in a complex sequence, like a series of interconnected bellows. The lungs, which are located midway between air sacs in terms of the flow of gas, are continuously ventilated in a single direction with freshly inspired air during both inspiration and expiration at the nostrils. Aspiration into the air sacs is produced by expansion of the chest and abdominal cavity. The sternum (breastbone) swings forward and downward, while the ribs and chest wall move laterally. Expiration is caused by compression of the air sacs by skeletal muscle.

As a consequence of the continual, unidirectional airflow, the lungs of birds are more completely ventilated than the lungs of mammals. The flow of gas and blood within the bird lung is carefully arranged to maximize gas exchange, which is far more efficient than in the mammalian lung: Himalayan geese have been observed not only to fly over human climbers struggling to reach the top of Mount Everest, but to honk as they do so. The ventilation of pigeons increases around 20-fold during flight, brought about by more rapid breathing and not by taking in more air at a breath. There is a precise synchrony between breathing and wing motion: the peak of expiration occurs at the downstroke of the wingbeat.

The respiratory system of birds is also used for communication through song. The “voice box” is the syrinx, a membranous structure at the lower end of the trachea. Sound is produced only when air flows outward across the syrinx. In canaries, notes or pulses are synchronous with chest movements; the trills, however, are made with a series of shallow breaths. The song of many small birds is of long duration relative to their breathing frequencies.

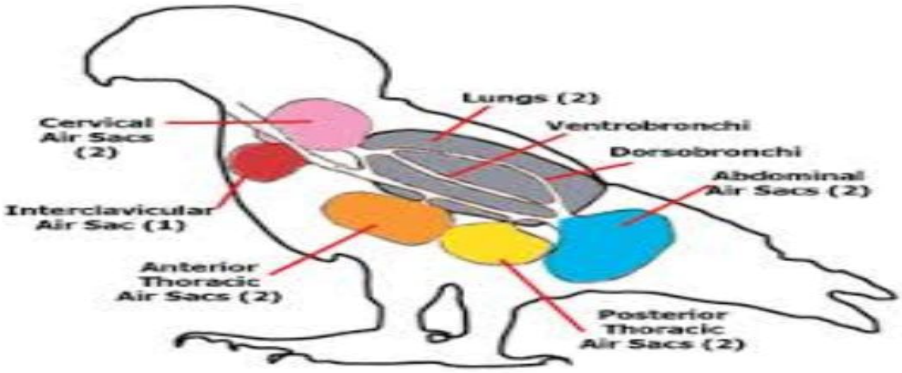


Fig. 9: Anatomy of a bird respiratory organs

Lung by Mammals

An increasing subdivision of the airways and the development of greater surface area at the exchange surfaces appear to be the general evolutionary trend among the higher vertebrates like man.

In the embryo, lungs develop as an outgrowth of the forward portion of the gut in varying degree among different vertebrate groups.

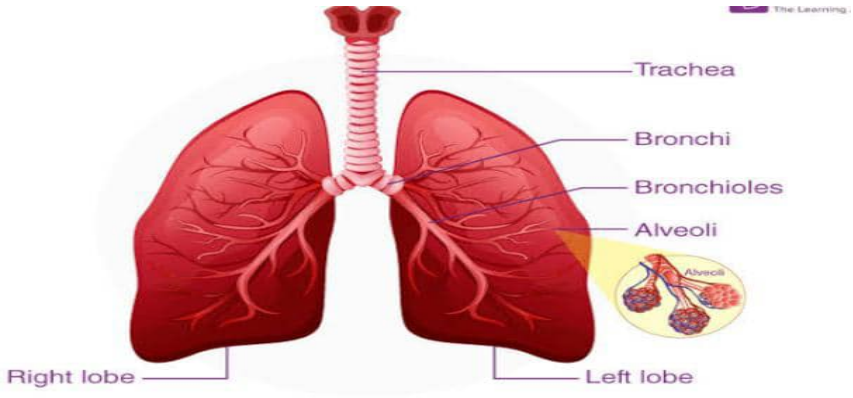


Fig. 10: The human lung

To provide the gas exchange necessary to support the elevated metabolic rate of mammals, mammalian lungs are subdivided internally. The trachea (wind pipe) branches into two bronchi. The presence of cartilagenous rings in the trachea and bronchi prevents them from collapsing when the air pressure in them is low. Each bronchus leads to the lung where it branches into small tubes called bronchioles. The alveoli are tiny balloon shaped air sacs at the end of the bronchioles richly supplied with blood capillaries and are sites where gaseous exchange takes place.

The repetitive subdivisions of the lung airways provide gas to the tiny alveoli (gas sacs) that form the functional gas-exchange surface area of the lungs.

Mechanism of Inspiration or Inhalation in Man

- The intercostal muscle contract
- The rib are moved upward and outward
- The diaphragm becomes flattened
- There is an increase in the volume of thoracic cavity
- Consequently air is drawn from the nostril to the trachea, bronchi, bronchioles, and finally to the alveoli.

Mechanism of Expiration or Exhalation in Man

- The intercostal muscle relax
- The ribs are moved downward and inward
- The diaphragm becomes dome shape
- There is decrease in the volume of thoracic cavity.
- Consequently air is drawn from the alveoli, bronchioles, bronchi, and trachea.
- Consequently air containing waste products like CO_2 and water vapour from inside the alveoli or lungs are forced out through bronchioles, bronchi, trachea, and finally to the exterior through the nos

MODE OF RESPIRATION IN PLANTS

Yes, like animals and humans, plants also breathe. Plants do require oxygen to respire, the process in return gives out carbon dioxide. Unlike humans and animals, plants do not possess any specialized structures for exchange of gases, however, they do possess stomata (found in leaves) and lenticels (found in stems) actively involved in the gaseous exchange. Leaves, stems and plant roots respire at a low pace compared to humans and animals.

We know that plants possess a specific ability to synthesize their own food through photosynthesis. Photosynthesis takes place in only those parts of the plants which have chlorophyll, the green plant parts. Photosynthesis is so evident that at times it seems to mask the respiratory process in plants. Respiration must not be mistaken for photosynthesis. Respiration occurs all through the day, but the photosynthesis process occurs in the daytime, in the presence of sunlight only. Consequently, respiration becomes evident at night time in plants. This is the reason we often hear people warn against sleeping under a tree during nighttime, as it may lead to suffocation due to excess amounts of carbon dioxide liberated by trees following respiration.

The Process of Respiration in Plants

Oxygen, carbon dioxide, and water vapour are released by simple diffusion process in plants. Oxygen diffuses into plants through the stomata and lenticels while CO_2 and water vapour diffuses out of the plant through the same opening. This is facilitated by the difference in the concentration gradient of these gases. The plant takes in oxygen mostly during the night and gives out carbon dioxide and water vapour during the day due to photosynthetic activities of the plant. Oxygen is the by-product of photosynthesis.

Respiration in Roots

Roots - the underground part of the plants, absorbs air from the air gaps/spaces found between the soil particles. Hence, absorbed oxygen through roots is utilized to liberate the energy that in the future, is used to transport salts and minerals from the soil.

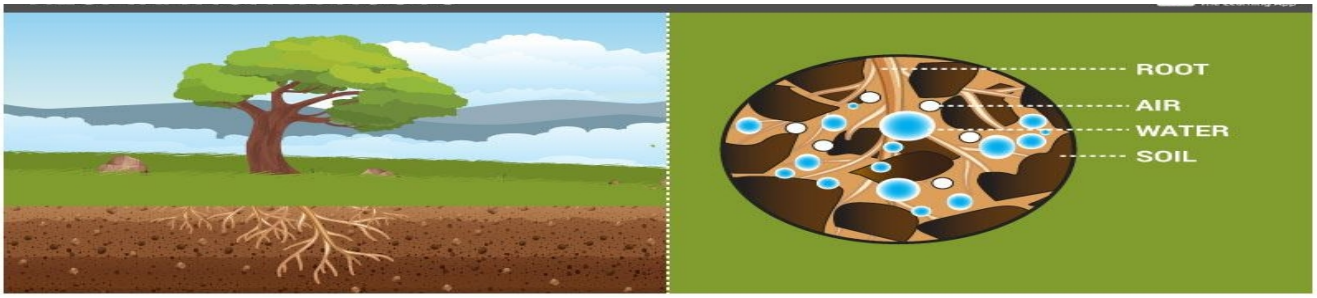


Fig. 12: Respiration by plant roots

Respiration in Stems



Fig. 13: Respiration in Stems

The air in case of stem diffuses into the stomata and moves through different parts of the cell to respire. During this stage, the carbon dioxide liberated is also diffused through the stomata. Lenticels are known to perform gaseous exchange in woody or higher plants.

Respiration In Leaves

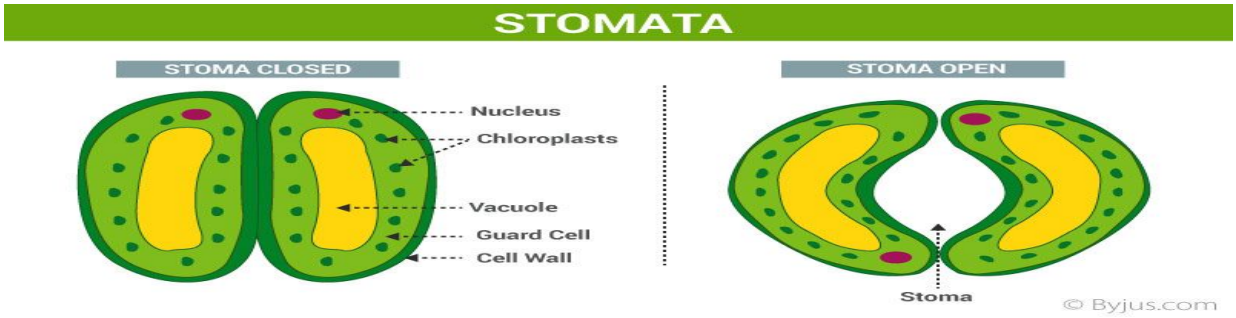


Fig. 14: Respiration by plant stomata

Leaves consist of tiny pores known as stomata. Gaseous exchange occurs through diffusion via stomata. Guard cells regulate each of the stomata. Exchange of gases occurs with the closing and opening of the stoma between the inferior of leaves and the atmosphere.

SUMMARY

Minute life-forms, such as protozoans, exchange oxygen and carbon dioxide across their entire surfaces. Multicellular organisms, in which diffusion distances are longer, generally resort to other strategies. Aquatic worms, for example, lengthen and flatten their bodies to refresh the external medium at their surfaces. Sessile sponges rely on the ebb and flow of ambient water. By contrast, the jellyfish, which can be quite large, has a low oxygen need because its content of organic matter is less than 1 percent and its metabolizing cells are located just beneath the surface, so that diffusing distances are small.

Organisms too large to satisfy their oxygen needs from the environment by diffusion are equipped with special respiratory structures in the form of gills, lungs, specialized areas of the intestine or pharynx (in certain fishes), or tracheae (air tubes penetrating the body wall, as in insects). Respiratory structures typically have an attenuated shape and a semi-permeable surface that is large in relation to the volume of the structure. Within them there is usually a circulation of body fluids (blood through the lungs, for example). Two sorts of pumping mechanisms are frequently encountered: one to renew the external oxygen-containing medium, the other to ensure circulation of the body fluids through the respiratory structure. In air-breathing vertebrates, alternately contracting sets of muscles create the pressure differences needed to expand or deflate the lungs, while the heart pumps blood through the respiratory surfaces within the lungs. Oxygenated blood returning to the heart is then pumped through the vascular system to the various tissues where the oxygen is consumed.